

Why You Should Deploy Immutable Infrastructure on the Cloud

Even as DevOps and cloud adoption have increased significantly in recent years, the use of immutable infrastructure has also become pretty popular. This article outlines the core benefits of deploying immutable infrastructure on the cloud.



Immutable infrastructure refers to a state where the infra components like VMs, servers and containers are never modified once deployed. In case any changes are required, the existing components are destroyed and replaced by new ones.

In the real world, we often make changes in security patching and while deploying a new application version, as well as OS-specific changes. All these changes are made ‘in-place’ or with the help of configuration management tools like Chef, Ansible, Puppet, or SaltStack. That makes the whole infrastructure unstable and prone to configuration drift. This is where the concept of immutable infrastructure comes in.

Under this approach, a golden (master) image of a server is created that has all the required software

stacks preconfigured. The same image is used to provision new servers. If any changes are required in the configuration of the servers, a new image is baked, tested, and rolled out across environments.

Some of the benefits of this approach are listed below.

- Creates consistency across environments (dev/stage/prod), as all the servers are provisioned with the help of a particular golden image.
- Makes infrastructure more stable because everything is immutable, as ‘in-place’ changes are not permitted. Additionally, eliminates the risk of configuration drift.
- Simplifies the deployment process because all the components are preconfigured on the server. Also

makes rollback safer since the same process used to deploy the new version is used to roll back to older versions.

- Makes debugging much faster during production issues since all the servers, provisioned using golden images, are properly versioned and well documented.
- Enhances security as the images are baked while keeping compliances in mind and validated after thorough testing.
- Finally, embraces Agile software deployment in practice by implementing DevOps principles. Terraform, Vagrant, Docker and Packer are some of the open source tools available for creating and managing immutable infrastructure on the cloud.